Night and Day: Diurnal Warming and Structural Transformation in India

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Abstract

This paper shows how decadal climate change shifts the composition of local economic activity through asymmetric diurnal warming—higher nighttime vs daytime temperatures—within Indian districts (1981–2011). Though both measures contract agricultural production and land-use, only warmer daytime temperatures raise local harvest prices. Warmer nights raise, warmer days lower the share of agricultural wage-labourers. Long differences show this divergence is rural. In towns, both warmer nights and days depress non-agricultural worker shares. These findings are consistent with a two-sector model of structural change where warmer days shock agricultural land and labour, but warmer nights are land-biased shocks.

Keywords: Climate Change, Structural Transformation, Agriculture, India
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Climate change poses a fundamental challenge to structural transformation—the reallocation of labour from agriculture to non-agricultural employment—a transition that underpins economic development and growth (Lewis 1954; Matsuyama 1992; Kongsamut, Rebelo, and Xie 2001; Herrendorf, Rogerson, and Valentinyi 2014). In developing economies, where large shares of the workforce remain employed in climate-sensitive agriculture, rising temperatures are a primary environmental driver of productivity and employment, explaining substantially greater variation in agricultural output than precipitation shocks (Schlenker and Roberts 2009; Dell, Jones, and Olken 2014).

Sectoral labour reallocation (out of agriculture) is therefore an important margin to mitigate the widening of the agricultural productivity gap in developing economies.[1] Yet, as a growing literature examines the role of migration (Jessoe, Manning, and Taylor 2017; Das and Sanyal 2026), informal employment (Pham 2026), trade barriers (Saxena 2024), and infrastructure (Dasgupta and Raghav 2024) in climate adaptation and labour reallocation, it remains unsettled whether warming itself accelerates or arrests the transition out of agriculture.

Within a fixed geographic context (Indian districts), two recent papers find opposite effects of rising temperatures on agricultural employment shares. While Colmer (2021) shows that higher temperatures push workers out of agriculture into manufacturing in the short run, Liu, Shamdasani, and Taraz (2023) find that long-run warming depresses local non-agricultural demand and traps labour within agriculture. Across countries, Nath (2025) similarly demonstrates that despite heat-induced agricultural productivity losses, subsistence food requirements ("the food problem" under Engel's law) can intensify agricultural specialisation.

This paper reconciles these diverging findings by decomposing warming along an economically and physically meaningful margin: the *time of day* at which warming occurs. The economic literature primarily measures warming unidimensionally—through daily mean temperatures, degree-day indices, or binned temperature thresholds (Schlenker, Hanemann, and C. 2006; Dell, Jones, and Olken 2014; Blakeslee and Fishman 2015; Somanathan et al. 2021)—compressing the 24-hour heat cycle into a single scalar. However, daily minimum temperatures ($T_{\min}$, occurring at night) and daily maximum temperatures ($T_{\max}$, occurring during the day) jointly govern both thermal *levels* and the diurnal temperature range (DTR $\equiv T_{\max} - T_{\min}$). Asymmetrically warmer nights can *compress* the DTR, while warmer days *expand* it—though both measures raise the levels.

Figure 1 illustrates the spatial variation that identifies these competing forces, plotting two-decade long-differenced changes in 10-year average growing-season nighttime ($\Delta T_{\min}$) and daytime ($\Delta T_{\max}$) temperatures between 1981–1991 and 2001–2011. Although both nighttime and daytime temperatures rose in *levels* across nearly the entire map,

[1] In India, agriculture contributes roughly 17% of total value added while employing roughly 42% of the workforce, implying a "productivity gap" of 25 percentage points (World Bank, 2025).

changes in the diurnal *range* exhibit stark spatial sorting. Following the post-1991 onset of asymmetric diurnal warming in South Asia (Karl et al. 1991, 1993; Easterling et al. 1997; Davy et al. 2016; Mall et al. 2021; IPCC 2021), nighttime warming outpaced daytime warming across the arid northwest and Indo-Gangetic plains (compressing the DTR), whereas daytime warming outpaced nighttime warming across the Deccan peninsula and coastal regions (expanding the DTR). This geography provides continuous cross-district variation in nighttime versus daytime warming across districts and towns.

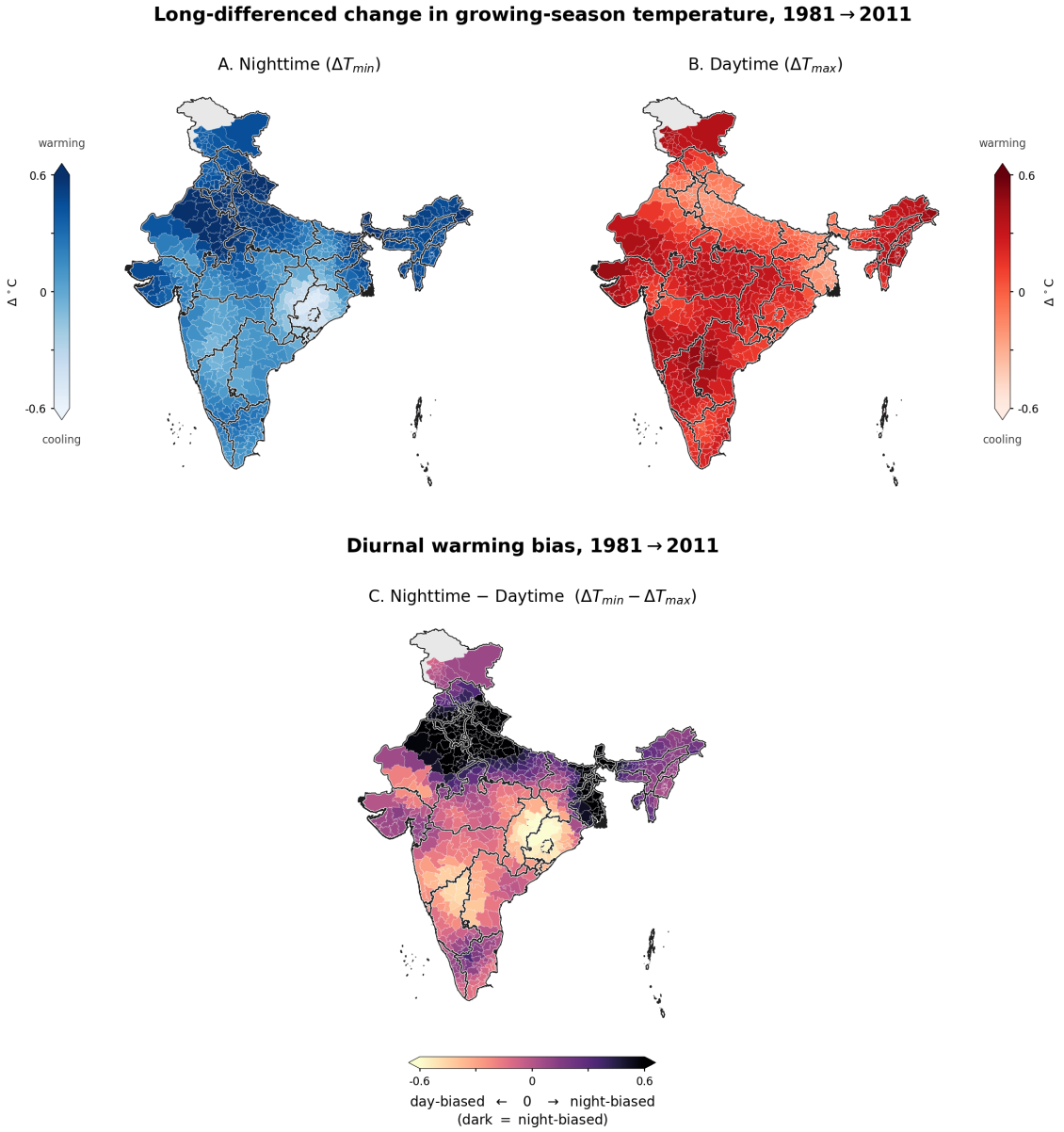
Why should nighttime and daytime warming move rural agricultural labour in opposite directions? The reason is a basic asymmetry in diurnal heat exposure: crops stand in the field around the clock, while agricultural labour works predominantly during daylight hours. Nighttime warming therefore acts primarily on the land factor, whereas daytime warming presses on both the crop and the outdoor worker, operating as a neutral (or labour-disaugmenting) productivity loss across the farm operation—formalised in Section 1.

To estimate how asymmetric diurnal warming reshapes local economic activity, I harmonise four decadal rounds of the Indian Census (1981–2011)—covering roughly 300 boundary-consistent districts and a balanced panel of 2,969 Census towns—with district agricultural statistics (ICRISAT) and urban employment surveys (NSS). I focus on labour reallocation *within* local labour markets in India, where caste-based insurance networks limit permanent long-run migration across districts (Munshi and Rosenzweig 2016) even as heat shocks induce short-run circulation (Das and Sanyal 2026). Using a long differences approach on the harmonised dataset, I show that nighttime and daytime warming move rural agricultural labour in opposite directions: holding the other margin constant, warmer nights ($\Delta T_{\min}$) *raise* and warmer days ($\Delta T_{\max}$) *lower* the share of full-time agricultural wage-labourers.

I document four core empirical findings. First, within the district agricultural economy (Table 1), warmer nights and warmer days symmetrically contract cultivated grain area but diverge in their effects on crop yields.[2] Under a 1°C rise on either margin, log cultivated area falls by an indistinguishable amount ($\beta_{\Delta T_{\min}} = -0.584$, $\beta_{\Delta T_{\max}} = -0.607$, $F = 0.03$), and agricultural field wages fall similarly (-0.193 and -0.212, $F = 0.02$). Yet cereal yields per hectare diverge significantly ($F = 4.56$, $p = 0.034$): log yields hold slightly positive under warmer nights ($+0.052$) as farms intensify labour on remaining land, whereas yields fall by 0.178 log points under warmer days. Over this two-decade horizon, regional trade arbitrages away local harvest-price wedges ($F = 1.81$); in medium-run decadal panel estimates (Online Appendix), however, daytime warming acts as a local supply shock that raises farm-gate harvest prices by $+0.130$ log points while nighttime warming leaves prices flat ($F = 13.34$).

[2]The core staple grain basket in the ICRISAT data comprises rice, wheat, maize, barley, sorghum, and millets.

FIGURE 1. The diurnal warming geography of India



Notes: Spatial distribution of warming nighttime and daytime temperatures in India. This map plots the change in decadal temperatures over the growing-season (June–Feb), by long-differencing the average between two endpoints: (2001,2011) − (1981,1991). IMD gridded climate is assigned to each district centroid by inverse-distance-squared weighting. Panels A and B map the change in the nighttime minimum (ΔT_{min}) and the change in daytime maximum (ΔT_{max})—darker where warming is stronger. Panel C maps the diurnal warming bias ($\Delta T_{min} - \Delta T_{max}$): darker where nights warmed faster (compressing DTR) and lighter where days warmed faster (widening DTR). Across the 639 districts with map coverage, nights outpaced days across the Indo-Gangetic north, while days outpaced nights across the peninsular Deccan (the estimation panel covers roughly 300 districts, Section 2).

Second, across districts (Appendix Table A3, Panel A), warmer nights raise and warmer days lower the share of full-time agricultural wage-labourers, with casual seasonal employment acting as the counter-flow. A 1°C rise in nighttime minimum temperatures ($\beta_{\Delta T_{\min}}$) *increases* the agricultural wage-labour share by $+7.5$ percentage points, drawing workers out of casual 'marginal' employment (-7.4 percentage points) and self-cultivation (-2.4 percentage points).[3] Conversely, a 1°C rise in daytime maximum temperatures ($\beta_{\Delta T_{\max}}$) *decreases* the agricultural wage-labour share by 6.2 percentage points and shifts $+4.9$ percentage points into the seasonal margin. The nighttime and daytime coefficients strongly reject equality on both agricultural wage-labour ($F = 26.07$, $p < 0.001$) and seasonal worker shares ($F = 12.34$, $p < 0.001$)—loading entirely on the diurnal range rather than the daily mean—while district non-agricultural worker shares remain flat ($F = 0.03$).

Third, decomposing districts by rural and urban residence (Appendix Table A3) shows that this labour tug-of-war is concentrated in the rural workforce ($F = 30.9$ on agricultural wage labour; $F = 11.4$ on seasonal work) and muted in the urban population ($F = 1.9$ and $F = 3.2$). Against a secular decline in the rural agricultural wage-labour share ($\Delta\bar{y}^{\text{rural}} = -0.046$), a 1°C warmer night pulls labour back into full-time farming ($+0.096$, reversing the net trend to $+0.050$), whereas a warmer day deepens the rural exit (-0.068, to -0.115).

Fourth, within the balanced panel of 2,969 Census towns (Table 2, Panel B), where agriculture is a minor sector, the night–day split on agricultural wage labour disappears ($F = 1.61$) as warming on either margin offsets the secular retreat from peri-urban farming. Instead, both warmer nights ($\beta_{\Delta T_{\min}} = -0.038$) and warmer days ($\beta_{\Delta T_{\max}} = -0.078$) unilaterally contract the town non-agricultural worker share ($F = 2.78$), stalling structural transformation at the urban margin.

Section 1 demonstrates that these findings are consistent with a two-sector general-equilibrium model of structural change in which agricultural land and labour are gross substitutes ($\sigma > 1$; Binswanger 1974; Hayami and Ruttan 1985) and warmer nights act as land-biased shocks while warmer days shock both land and labour (Bustos, Caprettini, and Ponticelli 2016; Barrett, Ortiz-Bobea, and Pham 2023). When local food markets are sufficiently integrated ($\chi < \chi^*$), a land-biased nighttime shock alters the optimal land-to-labour ratio and induces farms to intensify hired labour per hectare—buffering crop yields and drawing workers out of seasonal informality—whereas a neutral daytime shock shrinks the scale of production, reduces yields, raises medium-run harvest prices, and sheds full-time wage-labourers into the casual seasonal pool. In towns, where baseline agricultural employment is small, direct heat-capacity losses in informal outdoor work dominate under both margins and contract non-agricultural employment. Read

[3]The Census of India classifies 'marginal workers' as those employed for fewer than 183 calendar days in the year preceding enumeration—referred to throughout as casual 'seasonal' workers.

together, the results reveal a two-pronged arrest of structural transformation: warmer nights consolidate rural labour on the farm, while warmer days casualise the rural workforce and squeeze urban non-agricultural employment.

The rest of the paper proceeds as follows. Section 1 develops the conceptual framework. Section 2 describes the data and empirical approach. Section 3 presents the empirical results. Section 4 concludes.

1. Conceptual Framework

This section outlines a two-sector general-equilibrium framework that maps nighttime and daytime warming to agricultural production, crop yields, factor prices, and workforce reallocation. A detailed characterisation of the model and derivation of proofs is available in the Online Appendix.

What do nighttime and daytime temperatures measure? Nighttime minimum ($T_{\min}$) and daytime maximum ($T_{\max}$) temperatures are the extrema of the diurnal temperature range (DTR $\equiv T_{\max} - T_{\min}$). Holding the other constant, warmer nights compress the DTR from below and warmer days expand it at the upper tail. Because the pair $(T_{\min}, T_{\max})$ is a linear transformation of the daily mean $\frac{1}{2}(T_{\min} + T_{\max})$ and the range $(T_{\max} - T_{\min})$, entering $T_{\min}$ and $T_{\max}$ simultaneously spans both the thermal *level* and the diurnal *range*.

Night and day surface-air temperatures are also distinct economic shocks in agricultural production because *crops occupy fields around the clock, whereas field labour works predominantly during daylight hours*. Warmer nights act directly on the standing crop and soil moisture on cultivated land without exposing workers to daytime heat: humidity-adjusted (wet-bulb) nighttime warming raises cereal yields outright, the signature of land augmentation, while in dry-bulb surface-air temperatures this augmentation channel bundles with evaporative moisture stress on marginal land (Peng et al. 2004; Welch et al. 2010) (Appendix Remark A2). Warmer days distress *both* the crop and the outdoor worker, pressing human physical exertion against physiological limits and lowering overall productivity (Dell, Jones, and Olken 2012; Dunne, Stouffer, and John 2013; Somanathan et al. 2021). As a result, thermal levels govern the farm's extensive scale—contracting cultivated area under both margins—while the diurnal range governs intensive factor reallocation, crop yields, and medium-run harvest prices.

Technology, factor bias, and crop yields. A local economy (a district, or its rural, urban, or town workforce) produces an agricultural good Y_a at relative price p and a numeraire non-agricultural good $Y_n = A_n L_n^{\nu}$ ($\nu \in (0,1)$). Agricultural output combines field labour L_a and effective land $B(T_{\min})Z$ through a constant-elasticity-of-substitution (CES) production

function scaled by total factor productivity $A(T_{\max})$:

$$(1) \quad Y_a \;=\; A(T_{\max})\Big[\,\theta\, L_a^{\rho} + (1-\theta)\big(B(T_{\min})Z\big)^{\rho}\,\Big]^{1/\rho}, \qquad \rho \;\equiv\; \frac{\sigma-1}{\sigma}, \quad \theta \in (0,1),$$

where Z is physical cultivated area, $B(T_{\min})$ scales the productive services of cultivated land, and $\sigma > 0$ is the elasticity of substitution between labour and land. Grounded in the agricultural production literature (Binswanger 1974; Hayami and Ruttan 1985; Binswanger and Rosenzweig 1986), I maintain that labour and land are gross substitutes ($\sigma > 1$; Appendix Assumption A1). Daytime warming acts as a Hicks-neutral productivity loss ($A'(T_{\max}) < 0$, or equivalently a direct labour-disaugmenting shock $A'_L(T_{\max}) < 0$; Appendix Lemma A3), whereas nighttime warming acts as a *land-biased* shock ($B'(T_{\min}) \gtrless 0$, with $B'(T_{\min}) \geq 0$ as the intensive benchmark; Appendix Remark A2). A pure nighttime labour-fatigue shock cannot substitute for this land-bias channel: generating the observed nighttime rise in agricultural wage labour requires the land-biased term to dominate any nighttime labour-disaugmenting term (Appendix Remark A1).

Writing log deviations from an initial equilibrium with hats ($\hat{x} \equiv d\ln x$) and the equilibrium cost shares of labour and effective land as $s_L \in (0,1)$ and $s_Z = 1 - s_L$, the farm's first-order condition for field labour ($p\,\partial Y_a/\partial L_a = w$) yields the intensive labour-demand schedule (Appendix Lemma A1):

$$(2) \qquad \hat{L}_a - \hat{Z} \;=\; \hat{B} \;+\; \varepsilon_a\big(\hat{p} + \hat{A} - \hat{w}\big), \qquad \varepsilon_a \;\equiv\; \frac{\sigma}{s_Z} \;>\; 0.$$

Equation (2) is the core mechanism of the paper. The land-biased shock $\hat{B}$ enters farm labour demand with a *direct unit coefficient*: when warmer nights alter effective land services ($\hat{B} > 0$), a farm with $\sigma > 1$ reworks its factor mix and intensifies hired field labour per hectare ($\hat{L}_a - \hat{Z} > 0$), paralleling the factor-bias mechanism documented by Bustos, Caprettini, and Ponticelli (2016). By contrast, the neutral daytime shock $\hat{A} < 0$ enters only through the profitability block $\varepsilon_a(\hat{p} + \hat{A} - \hat{w})$: it leaves the relative factor mix intact at given prices and shrinks the scale of operation, depressing labour demand and the field wage. Under Cobb–Douglas ($\sigma = 1 \implies \varepsilon_a = 1/s_Z$), the land-biased shock folds into neutral TFP ($\hat{A} + s_Z\hat{B}$) and the diurnal labour distinction vanishes (Appendix Lemma A2).

On the extensive margin, farms choose cultivated area Z against a convex cultivation cost $C(Z)$ subject to climate stress shifters $\psi_T\, dT \geq 0$ as marginal parcels fall below break-even viability (Appendix Proposition A1). Both nighttime and daytime warming contract cultivated area ($\hat{Z} < 0$), loading on the diurnal mean. Yet because log crop yield per hectare ($y_a \equiv Y_a/Z$) satisfies $\hat{y}_a = \hat{Y}_a - \hat{Z} = \hat{A} + s_L(\hat{L}_a - \hat{Z}) + s_Z\hat{B}$, nighttime labour intensification ($\hat{L}_a - \hat{Z} > 0$) buffers and raises crop yields on remaining land relative to daytime labour shedding, generating a positive night-minus-day yield wedge

($\hat{y}_{a,T_{\min}} > \hat{y}_{a,T_{\max}}$; Appendix Proposition A2).

The seasonal employment 'sink'. The local workforce $\bar{L}$ is partitioned across four mutually exclusive Census categories whose shares sum to unity: $\ell_a + \ell_c + \ell_n + \ell_s = 1$, denoting full-time agricultural wage labour (ℓ_a), self-cultivation (ℓ_c), full-time non-agricultural work (ℓ_n), and casual *seasonal* (marginal) work (ℓ_s). Casual seasonal workers draw heterogeneous reservation wages and supply full-time market labour with wage elasticity $\eta_s \geq 0$, while marginal self-cultivators hire out into the wage market when the return on their own marginal parcel falls below w. Combining seasonal supply with non-agricultural labour demand ($\hat{L}_n = \varepsilon_n(\hat{A}_n - \hat{w})$, where $\varepsilon_n \equiv \frac{1}{1-\nu}$) yields an upward-sloping inverse labour supply schedule facing agriculture, $\hat{w} = (\phi_a/\eta_a)\hat{L}_a$, where $\eta_a \equiv \eta_s + \phi_n\varepsilon_n > 0$ is the effective labour supply elasticity and ϕ_a, ϕ_n are initial market-labour shares. When district non-agricultural shares remain flat ($d\ell_n \approx 0$), workforce adding-up requires $d\ell_c + d\ell_s = -d\ell_a$ (Appendix Lemma A4): under warmer nights ($\hat{L}_a > 0$), expanding farm wage labour draws workers out of marginal self-cultivation and the seasonal pool; under warmer days ($\hat{L}_a < 0$), displaced farm wage workers are shed back into the elastic seasonal buffer.

Market closure, harvest prices, and the openness threshold. Households have Stone–Geary preferences with a subsistence food floor $\bar{c}_a > 0$. In a closed local economy, market clearing implies $\hat{p}_{\text{closed}} = \hat{Y}_n - \mathcal{M}\hat{Y}_a$, where $\mathcal{M} \equiv \frac{Y_a}{Y_a - \bar{c}_a\bar{L}} > 1$ is the subsistence multiplier. Nesting open and closed regional food markets via a reduced-form closure parameter $\chi \in [0,1]$ such that $\hat{p} = \chi(\hat{Y}_n - \mathcal{M}\hat{Y}_a)$ (where $\chi = 0$ is a price-taking open district and $\chi = 1$ is autarky), equilibrium agricultural labour (at fixed area, derived in the Online Appendix) is:

$$\hat{L}_a \;=\; \frac{1 - \sigma\chi\mathcal{M}}{D(\chi)}\,\hat{B} \;+\; \frac{\varepsilon_a(1 - \chi\mathcal{M})}{D(\chi)}\,\hat{A}, \tag{3}$$

where $D(\chi) > 0$. In an open economy ($\chi = 0 \implies \hat{p} = 0$), equation (3) reduces to $\hat{L}_a = \tau_a(\hat{B} + \varepsilon_a\hat{A})$ with incidence ratio $\tau_a \equiv \frac{\eta_a}{\eta_a + \phi_a\varepsilon_a} \in (0,1)$, so warmer nights raise ($\hat{B} > 0$) and warmer days lower ($\hat{A} < 0$) agricultural wage labour (Appendix Proposition A3). In a fully closed subsistence economy ($\chi = 1$), by contrast, $\mathcal{M} > 1$ and $\sigma > 1$ turn both numerators in (3) negative, *reversing* both signs (Appendix Proposition A4). Generating the empirically observed orientation—warmer nights raising and warmer days lowering agricultural wage labour—requires local food markets to be sufficiently open: $\chi < \chi^* \equiv \frac{1}{\sigma\mathcal{M}}$ (Appendix Proposition A5). Over the two-decade long difference, regional trade arbitrage drives $\chi \to 0$, leaving long-run local harvest prices flat under both margins (Table 1); in the medium-run decadal panel, partial local closure ($0 < \chi < \chi^*$) allows the daytime output loss on inframarginal marketed grain to raise local harvest prices

while nighttime abandonment of subsistence marginal plots leaves marketed prices flat (Appendix Remark A3).

Rural–urban amplitude and the town non-agricultural drag. Because both rural and urban district populations exhibit a positive nighttime agricultural labour response—the binding condition for $\chi < \chi^*$—they share the same open-economy sign regime but differ in amplitude (Appendix Proposition A6). Since the share response $d\ell_a = \ell_a \hat{L}_a$ scales with the baseline agricultural employment share ℓ_a and the seasonal supply elasticity η_s (which raises τ_a toward 1), the diurnal divergence registers sharply in rural district populations and attenuates in urban populations. In Census towns, where by construction at least 75% of male workers are employed outside agriculture (ℓ_a and ϕ_a small), the within-agriculture factor-bias split recedes and the direct thermal sensitivity of outdoor informal non-agricultural work ($\hat{A}_n < 0$ under both daytime heat strain and nighttime recovery loss) dominates, contracting town non-agricultural employment shares under both warmer nights and warmer days (Appendix Proposition A7).

2. Data and Empirical Approach

2.1. Data Sources

I harmonise four datasets to link climate, agricultural production and structural transformation: (1) daily gridded weather data from the IMD, (2) district agricultural data (ICRISAT), (3) district and town level tables from the Indian Census, and (4) urban household-level employment data from the NSS. These datasets span three decades from 1981 to 2011 (Census, IMD, ICRISAT), with a complementary window from the NSS (1987, 1999, 2009).

Indian Meteorological Department. Gridded public data were obtained from the Indian Meteorological Department (IMD) on daily surface-air temperatures (1° resolution) and daily precipitation (0.25° resolution) between 1951–2023. To measure persistent climate shifts, I average daily minimum ($T_{\min}$) and maximum ($T_{\max}$) surface-air temperatures over 10 years of growing seasons. These seasons span the months of June–February in India, which hold the cropping cycles for rice in the monsoon (kharif) and wheat in the winter (rabi) seasons. As robustness measures, I also construct wet-bulb temperature approximations (Stull 2011) and degree days (DDs) (Blakeslee and Fishman 2015; Schlenker, Hanemann, and C. 2006).

Census of India. I utilise a digitised panel from raw image scans of the District Census Handbooks (DCHBs) of India.[4] Specifically, the dataset covers the Primary Census Abstract (PCA) of each handbook, covering the Census universe of Indian districts and towns from 1981 to 2011.[5] I construct a panel of 2,969 towns that appear consistently across all four decades, so that the analysis reflects intensive-margin adjustments instead of how urban areas are classified. For comparability over time, occupations are aggregated into four mutually exclusive categories: cultivators (landowners), agricultural labourers (wage workers), non-agricultural workers, and marginal workers. The "marginal" category cannot be assigned cleanly to agriculture or non-agriculture, but it lets me track the contractual *quality* of work. The 2001 Census enumerated marginal work more completely than 1991; the headline results are identified off the consistently defined main-worker categories, and a per-capita denomination immune to the change reproduces them (online appendix). The district panel covers roughly 300 districts across twenty major states as of the 2011 Census with harmonised administrative boundaries.

ICRISAT. The ICRISAT District Level Database (DLD) tracks agricultural inputs and production in India from 1966 to the present. I harmonise districts to 2011-Census consistent boundaries, constructing an annual panel of cultivated area, production, yields and harvest prices for key grains: rice, wheat, maize, sorghum, barley and millets. I construct the mean of these measures over 10 years preceding each Census round, e.g. over the years 1971–1980 for the 1981 Census. Thus, the results from the temperature regressions compare contemporaneous horizons of 10-year windows with climate data. The DLD covers staple grains only; horticulture and cash crops—which have grown rapidly and differ in labour intensity, irrigation, and market structure—lie outside it, so the agricultural-economy results speak to staple-grain production and the broader composition of Indian agriculture is left to future work. The Census labour reallocation, however, survives this composition shift: it is strongest in districts that stayed in staple grains and muted where agriculture diversified into high-value crops (online appendix).

National Sample Surveys. I use three rounds of the National Sample Survey (NSS) Employment and Unemployment Schedules (1987, 1999, and 2009), harmonised by IPUMS (Ruggles et al. 2024). While the Census measures the volume of labour reallocation, it lacks a granular view of the destination sectors within non-agriculture under its 1981 definitions. Restricting the NSS sample to urban households serves as an independent benchmark to the Census urban and town-level data. The NSS urban sampling frame is separate from Census towns (it retains more peri-urban agriculture), and provides

[4]The data harmonisation builds on data assembled for Blakeslee et al. (2023).

[5]A Census town meets three criteria: (1) a minimum population of 4,000, (2) at least 75% of the male working population in non-agricultural activity, and (3) a population density above 400 persons per sq. km.

complementary evidence of the divergence in agricultural wage-labour shares. The data additionally shed light on the destination of heat-displaced labour—casual, informal work in manufacturing or services.

2.2. Empirical Approach

The identification takes two approaches: a fixed effects panel estimation (4), and a long differences approach (5), consistent with Dell, Jones, and Olken (2014), Burke and Emerick (2016) and Liu, Shamdasani, and Taraz (2023). Let Y_{irt} be the outcome of interest (e.g., share of workers in agriculture) for district or town i, within state r, observed at time t. $\mathbf{T}_{irt}$ denotes a vector of temperature measures, either degree days $\mathbf{T} = \{DDs\}$ or the diurnal surface-air pair $\mathbf{T} = \{T_{min}, T_{max}\}$. All specifications control for total precipitation $\mathbf{P}_{it}$ and fit a state-specific linear time trend.

$$Y_{irt} = \beta \mathbf{T}_{irt} + \phi \mathbf{P}_{irt} + \mu_i + \lambda_t + \gamma_r \cdot t + \varepsilon_{irt} \tag{4}$$

$$\Delta Y_{ir} = \beta \Delta \mathbf{T}_{ir} + \phi \Delta \mathbf{P}_{ir} + \gamma_r + \varepsilon_{ir} \tag{5}$$

For temporal parity, I construct the gridded temperature and rainfall measures as a moving average over the growing seasons of the preceding ten years. Decadal climate is the inverse-distance-squared–weighted average of IMD grid points surrounding the district (within 200 km) or town (within 100 km) centroid—radii chosen to match the distance at which the residual spatial correlation of the outcomes decays (roughly 150–200 km for districts and 100 km for towns). Because the weights fall off as the squared distance, grid points beyond roughly 100 km contribute negligibly, so the assignment is insensitive to the exact cutoff.

Equation 4 is a district- or town-decade panel with two-way fixed effects and a state-specific linear trend; because the climate regressor is itself a ten-year moving average, I read its coefficients as the *decadal* response of structural transformation to heat. Equation 5 averages the outcome Y and covariates X over two endpoints—1981–91 and 2001–11—and regresses $\Delta Y = Y_2 - Y_1$ on $\Delta X = X_2 - X_1$. This long-differenced estimator recovers the effect of a sustained increase in $\mathbf{T}$ on the outcome trend ΔY, measured against its secular trend $\Delta \bar{Y}$ over the sample. Both estimators use the same decadally smoothed climate; the long difference adds one further decade of horizon and absorbs unobserved state-level trajectories $(\gamma_r \cdot t)$. I treat it as a robustness design—confirming that the decadal-panel result survives a different fixed-effects and averaging structure

that strips decade-common shocks.[6]

I enter T_{min} and T_{max} simultaneously and read each coefficient as the partial effect of one margin holding the other—and rainfall—constant. Inference faces two distinct dependencies. The first is the decadal *persistence* of the climate regressor (a ten-year moving average), which inflates rejection rates unless the estimator accounts for within-unit serial correlation. This choice is disciplined by Monte Carlo simulations (online appendix), where I regress the key outcomes on a placebo climate series constructed as a random permutation of the true data across districts in a given decade. I select the preferred panel estimator for equation 4 as the specification that rejects the null in roughly 5% of simulations. I find that unadjusted and heteroskedasticity-robust standard errors over-reject sharply, whereas clustering by district with state-specific linear trends restores the nominal 5% size.

The second dependency is spatial—a persistent, strongly associated feature of Indian diurnal-temperature data (Bhattacharjee and Bose 2024); because the permutation reassigns whole series across units it breaks the regressor's spatial structure, so the Monte Carlo does not speak to spatial correlation, which I address separately with Conley spatial-HAC standard errors (Conley 1999) reported alongside the clustered errors on the panel exhibits. The online appendix sweeps the Conley radius to 200 km: the night/day equality rejection is stable at every radius ($F = 12.4$ at 200 km), with the daytime labour coefficient thinning to the ten-percent level at the widest radius.

The simultaneous estimator addresses two identification challenges. First, because T_{min} and T_{max} are correlated ($\rho \approx 0.78$ across districts), entering either alone would induce omitted-variable bias; entering them together isolates their partial effects and identifies a narrowing versus widening of the local diurnal range. The district and decade fixed effects and state trends absorb most of this co-movement, so the *estimating* variation is close to orthogonal (within-transformed VIF ≈ 1; the online appendix) and the partial signs are well-identified. By Frisch–Waugh–Lovell, the coefficients are determined exactly from this residual variation, so the within correlation governs whether the two margins can be separated—and within, they are all but orthogonal. Absorbing state-specific trends accounts for time-varying state policies and compares districts and towns within their local economic geography under similar baseline climate trajectories. Correlograms of the residuals confirm that the spatial correlation of these heat shocks in the key outcomes decays within roughly 100 km for towns and 150–200 km for districts.

[6] Lemoine (2021) shows that long-difference and panel estimators of climate impacts are identified by overlapping variation in the realised temperature series (here decadally averaged), so a long difference need not isolate differential rates of climate change or a separate long-run adaptation response, and agreement between the two designs reflects robustness to the estimating structure. Identification here therefore rests on the within-unit, simultaneous T_{min}-versus-T_{max} contrast in the decadal panel; the ten-year smoothing and the long difference serve as robustness.

3. Results

Figure 2 previews the district-level results in one place: a unilateral contraction in cultivated area, a divergence in cereal yields, and the same divergence—mirrored in sign and magnitude—in agricultural wage-labour and seasonal work. Tables 1 and 2 present the long-differenced estimates behind this and the remaining two findings, in the same order as below: the district agricultural economy, district labour reallocation, the rural–urban split, and the town non-agricultural drag.

The district agricultural economy. Table 1 estimates the two-decade long-difference response (1981–1991 to 2001–2011) of the district grain economy—rice, wheat, maize, sorghum, barley, and millets—to a unit change in nighttime ($\Delta T_{\min}$) and daytime ($\Delta T_{\max}$) warming trends during growing seasons.

On the extensive margin, both temperature measures contract agricultural land use for grain cultivation: a 1°C rise in $\Delta T_{\min}$ or $\Delta T_{\max}$ reduces log cultivated area by 0.584 (SE $=$ 0.165) and 0.607 (SE $=$ 0.220) log points, respectively (column 1). The coefficients fail to reject pairwise equality ($F = 0.03$, $p = 0.859$). Simultaneously, total grain production falls by 0.289 log points (SE $=$ 0.193) under warmer nights and 0.424 log points (SE $=$ 0.246) under warmer days ($F = 0.66$, column 2).

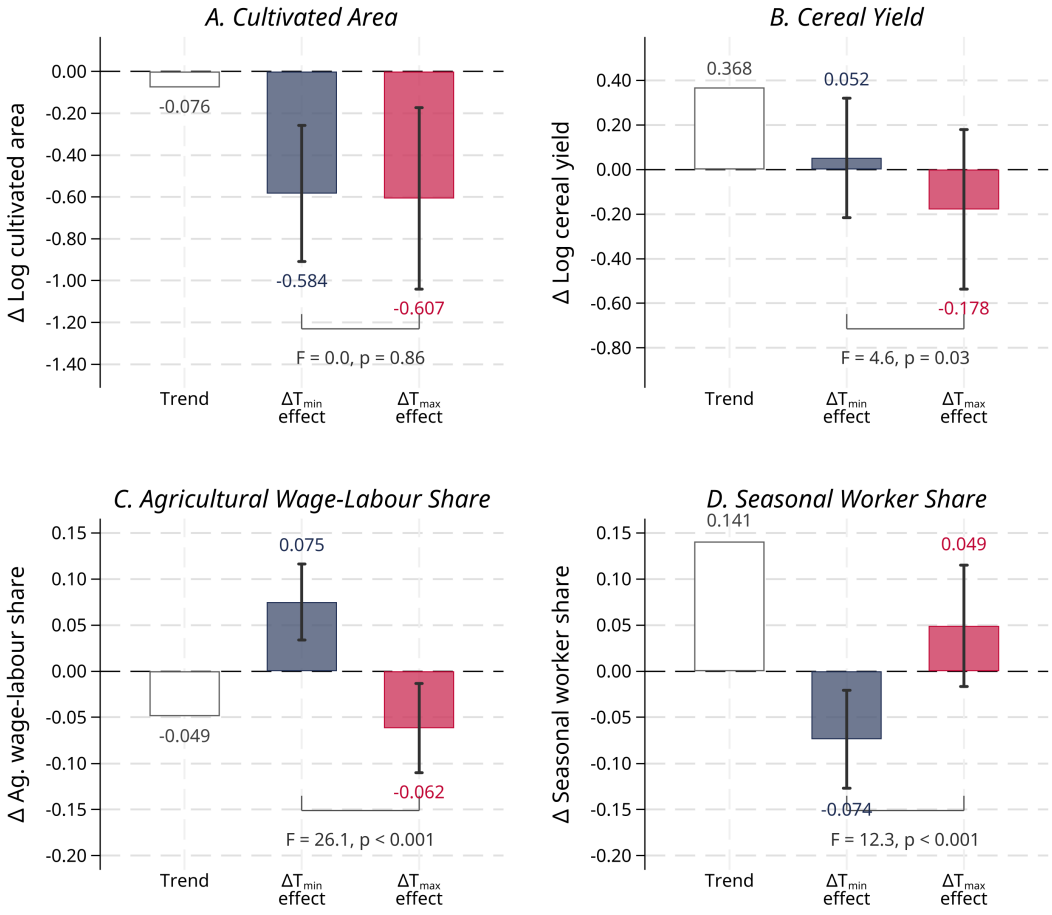
Although both nighttime and daytime warming significantly cut down on cultivated area, the effects on grain production yields are insignificant. Despite identical contractions in cultivated area, log cereal yields hold slightly positive under warmer nights ($+0.052$, SE $=$ 0.136) while falling by 0.178 log points (SE $=$ 0.182) under warmer days. Within yields, the effects of nighttime and daytime warming diverge as the two coefficients reject pairwise equality ($F = 4.56$, $p = 0.03$). These results together suggest adaptation from labour, the other input to the agricultural production function.

The conceptual framework (Section 1) rationalizes the diverging effects on yields through an underlying factor bias of either heat shock, combined with the gross substitutability ($\sigma > 1$) of land and labour in agricultural production. Because warmer nights act as a land-biased shock, farms intensify labour on remaining cultivated land which buffers the yield per hectare, while warmer days shock both land and outdoor labour, shedding workers and compounding the yield loss.[7]

Finally, over the two-decade horizon, there is no significant night-day divergence on the factor prices: neither in farm–gate harvest prices ($F = 1.81$, column 3), nor in field wages ($F = 0.02$, column 4). This result is consistent with the open-economy case of the

[7] In the Online Appendix, humidity-adjusted (wet-bulb) temperatures net out dry-air evaporative stress on marginal parcels and isolate the intensive land-augmentation channel directly, raising nighttime cereal yields outright ($+0.15$ in the decadal panel and $+0.36$ in the long difference). Crop-by-crop estimates also show that the cereal index averages over opposing grain responses, with heat-tolerant coarse cereals gaining under warmer nights while wheat loses.

FIGURE 2. Long-differenced estimates of diurnal warming on the district agricultural economy and labour reallocation



Notes: Long-difference estimates (1981–2011) of growing-season nighttime (T_{min}) and daytime (T_{max}) surface-air warming on the district agricultural economy (Panels A–B, ICRISAT) and district labour reallocation (Panels C–D, Census), the coefficients behind Tables 1 and 2. In each panel, the white bar plots the two-decade secular trend $\Delta\bar{y}$; the coloured bars plot the raw 1°C nighttime (ΔT_{min}, navy) and daytime (ΔT_{max}, cranberry) coefficients themselves, each with a 95% confidence interval (spike), not shifted by the trend—the implied net trajectory is the trend plus the coefficient. The brace reports the night-versus-day equality test ($\Delta T_{min} = \Delta T_{max}$). Panels C and D share a common vertical scale, so the near-mirrored night and day coefficients across agricultural wage-labour and seasonal work are directly comparable. All specifications control for the long-run change in precipitation and include state fixed effects; standard errors are clustered at the district level.

TABLE 1. Diurnal warming and the district agricultural economy, 1981–2011

	Δ Log Levels (All Grains)				
	Cultivated Area	Production	Harvest Price Index	Field Wage	Yield Index
	(1)	(2)	(3)	(4)	(5)
Panel A. Two-decade long difference, 1981–2011 (state FE)					
ΔT_{min}	-0.584	-0.289	-0.045	-0.193	0.052
	(0.165)	(0.193)	(0.049)	(0.104)	(0.136)
	[0.160]	[0.187]	[0.051]	[0.105]	[0.132]
ΔT_{max}	-0.607	-0.424	0.061	-0.212	-0.178
	(0.220)	(0.246)	(0.075)	(0.123)	(0.182)
	[0.214]	[0.239]	[0.073]	[0.115]	[0.176]
F-stat ($\Delta T_{min} = \Delta T_{max}$)	0.03	0.66	1.81	0.02	4.56
p-value	0.859	0.418	0.179	0.894	0.034
Dep. Var. Mean	-0.08	0.36	1.36	1.92	0.37
Observations	311	311	294	259	303
Rainfall Control	Y	Y	Y	Y	Y
Panel B. Decadal panel, 1981–2011 (district & decade FE, state trend)					
ΔT_{min}	-0.218	-0.145	-0.042	-0.038	0.051
	(0.046)	(0.064)	(0.031)	(0.060)	(0.040)
	[0.046]	[0.064]	[0.031]	[0.060]	[0.040]
ΔT_{max}	-0.225	-0.221	0.130	-0.119	-0.046
	(0.056)	(0.064)	(0.034)	(0.070)	(0.047)
	[0.056]	[0.064]	[0.034]	[0.068]	[0.047]
F-stat ($\Delta T_{min} = \Delta T_{max}$)	0.01	0.80	13.34	0.74	2.63
p-value	0.905	0.373	<0.001	0.391	0.106
Dep. Var. Mean	5.45	5.77	5.66	2.86	7.14
Observations	1,243	1,243	1,191	1,115	1,211
Rainfall Control	Y	Y	Y	Y	Y

Notes: Each column reports separate regressions of the outcome on growing-season (June–February) surface-air minimum (ΔT_{min}) and maximum (ΔT_{max}) temperature and total precipitation, on 2011 Census-consistent district boundaries. Outcomes are log cultivated area, log grain production, a log harvest-price index, the log agricultural field wage, and an area-weighted cereal yield index (rice, wheat, barley, maize, millets, sorghum) from the ICRISAT district data. Panel A is the two-decade long difference (1981–1991 to 2001–2011 decadal averages) with state fixed effects. Panel B is the decadal panel (1981–2011) with district and decade fixed effects and a state-specific linear trend; its medium-run harvest-price response ($F = 13.34$, column 3) is the over-identifying restriction discussed in Section 1. Standard errors clustered by district are in parentheses; Conley spatial HAC standard errors (50 km) are in brackets. The F-statistic and its p-value test the equality of the nighttime and daytime coefficients within a column: cultivated area and the field wage fall by the same amount under either margin in both panels; the harvest price separates only in the decadal panel, and the long-difference yield index diverges, holding under warmer nights and falling under warmer days.

theoretical model ($\chi \to 0$), where regional trade arbitrages away any price gaps in the long run. However, the decadal panel estimates (Online Appendix) show a divergence in local farm–gate prices, suggesting markets are not fully integrated in the medium-run ($0 < \chi < \chi^*$).

District labour reallocation. Appendix Table A3, Panel A, shows the effects of asymmetric diurnal warming on labour reallocation using a balanced panel of 293 Census districts (1981–2011). Warmer nights and days reallocate labour in opposite directions, between full-time agricultural wage labour and informal seasonal work.

A 1°C rise in nighttime minimum temperature (ΔT_{min}) *raises* the agricultural wage-labour share by 0.075 (7.5 percentage points, SE = 0.021, column 1), drawing workers primarily out of the casual seasonal margin (-0.074, SE = 0.027, column 3) along with a modest reduction in self-cultivation (-0.024, SE = 0.031, column 2). Conversely, a 1°C rise in daytime maximum temperature (ΔT_{max}) *lowers* the agricultural wage-labour share by 0.062 (SE = 0.025, column 1) and shifts $+0.049$ (SE = 0.033, column 3) toward the seasonal pool, with self-cultivation virtually unchanged (-0.008, SE = 0.038).

The nighttime and daytime coefficients reject pairwise equality on both the agricultural wage-labour share ($F = 26.07$, $p < 0.001$) and the seasonal worker share ($F = 12.34$, $p < 0.001$). Equivalently, when reparameterised into the diurnal mean 'level' and diurnal temperature 'range' through a linear transformation of the coefficients (Online Appendix), the mean has no significant effect on either agricultural or seasonal labour shares, while the range drives the entire reallocation. Meanwhile, the district non-agricultural worker share ($\Delta T_{\text{min}} : +0.026$, $\Delta T_{\text{max}} : +0.020$; $F = 0.03$, column 4) and log total district workers ($F = 0.83$, column 5) show no significant divergence.

These results from Panel A indicate diurnal warming churns the district workforce between full-time farm wage work and seasonal, informal employment. Warmer nights consolidate and commercialize employment in agriculture, warmer days casualize it.

Rural versus urban populations within districts. Table 2, Panel A, reports the rural-workforce estimates directly; Appendix Table A3 adds the urban split (Panel C) alongside the same rural numbers (Panel B) for direct comparison. For each population and margin, the implied net trajectory is the two-decade secular trend $\Delta\bar{y}$ plus the warming coefficient ($\Delta\bar{y} + \beta_{\Delta T_{\text{min}}}$ at night, $\Delta\bar{y} + \beta_{\Delta T_{\text{max}}}$ by day).

In the rural workforce, the agricultural wage-labour share declined by -0.046 (4.6 percentage points) over these two decades, and diurnal warming bends this secular trend sharply in opposite directions ($F = 30.9$, $p < 0.001$): a 1°C warmer night raises the rural agricultural wage-labour share by $+0.096$, reversing the -0.046 secular decline into a $+0.050$ net expansion, whereas a 1°C warmer day lowers the rural share by -0.068, deepening the decline to -0.115. The rural seasonal workforce shows the exact mirror

TABLE 2. Long-difference estimates of diurnal warming on district and town labour reallocation, 1981–2011

	Δ Share of Workers				
	Ag. Labour	Cultivator	Seasonal	Non-Agriculture	Δ Log Workers
	(1)	(2)	(3)	(4)	(5)
Panel A. Census districts, rural workforce (N=291)					
ΔT_{min}	0.096	-0.035	-0.064	0.007	0.211
	(0.023)	(0.031)	(0.029)	(0.020)	(0.231)
	[0.023]	[0.030]	[0.027]	[0.019]	[0.221]
ΔT_{max}	-0.068	0.011	0.056	-0.001	0.025
	(0.027)	(0.041)	(0.036)	(0.029)	(0.259)
	[0.028]	[0.040]	[0.034]	[0.028]	[0.249]
F-stat ($\Delta T_{min} = \Delta T_{max}$)	30.87	1.24	11.36	0.07	0.48
p-value	<0.001	0.266	<0.001	0.794	0.489
Dep. Var. Mean	-0.05	-0.15	0.15	0.04	0.06
Observations	291	291	291	291	293
Panel B. Census towns, balanced panel (N=2,969)					
ΔT_{min}	0.063	-0.009	-0.016	-0.038	0.272
	(0.012)	(0.011)	(0.012)	(0.018)	(0.169)
	[0.012]	[0.011]	[0.012]	[0.019]	[0.160]
ΔT_{max}	0.041	0.022	0.017	-0.078	0.241
	(0.016)	(0.014)	(0.014)	(0.022)	(0.171)
	[0.017]	[0.015]	[0.016]	[0.024]	[0.165]
F-stat ($\Delta T_{min} = \Delta T_{max}$)	1.61	3.38	4.06	2.78	0.11
p-value	0.205	0.067	0.044	0.096	0.738
Dep. Var. Mean	-0.05	-0.06	0.10	0.01	0.55
Observations	2,969	2,969	2,969	2,969	2,969
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Each column reports a separate long-difference regression (1981–1991 to 2001–2011 decadal averages) of the change in a Census worker-category share (columns 1–4, which sum to unity) or in log total workers (column 5) on the changes in growing-season ΔT_{min} and ΔT_{max}, controlling for the change in precipitation, with state fixed effects. Panel A covers the rural workforce of 2011 Census-consistent districts—the sharpest of the within-district residence splits (Appendix Table A3 reports the total and urban splits alongside it); Panel B covers the balanced panel of Census towns observed in every Census round from 1981 to 2011. Standard errors clustered by district are in parentheses; Conley spatial HAC standard errors (50 km) are in brackets. The F-statistic and its p-value test the equality of the nighttime and daytime coefficients within a column. In the rural district workforce (Panel A) the test rejects for agricultural wage labour and for seasonal work, with opposite signs on the two margins; in towns (Panel B) the two margins move agricultural labour in the same direction and both lower the non-agricultural share.

image ($F = 11.4$, $p < 0.001$): against a secular rise of $+0.154$ in the rural seasonal share, warmer nights pull workers out of seasonal informality ($\beta_{\Delta T_{\min}} = -0.064$, lowering the net trend to $+0.089$), while warmer days push displaced farmworkers into the seasonal pool ($\beta_{\Delta T_{\max}} = +0.056$, lifting the net trend to $+0.210$, matching the $+0.032$ decadal panel inflow in the Online Appendix).

In the urban populations of the very same districts, both divergences attenuate sharply. A warmer night lifts the urban agricultural wage-labour share by $+0.034$—erasing the -0.035 urban secular decline to -0.001—while a warmer day leaves it unchanged ($\beta_{\Delta T_{\max}} = +0.006$, taking the trend to -0.029; $F = 1.9$, $p = 0.17$). Similarly, the urban seasonal response is muted ($\Delta\bar{y} = +0.097$, shifting to $+0.069$ at night and $+0.103$ by day; $F = 3.2$, $p = 0.07$). As shown in Section 1 (Appendix Proposition A6), this rural–urban gradient is an amplitude effect: because the factor-bias mechanism operates inside agriculture, its footprint in local workforce shares scales with baseline agricultural employment (ℓ_a) and the depth of the rural seasonal buffer (η_s).

Towns and the non-agricultural drag. Table 2, Panel B, turns to the balanced panel of 2,969 Census towns observed in every round from 1981 to 2011. Because Census towns are defined by having at least 75% of male workers outside agriculture, the within-agriculture factor-bias tug-of-war recedes and a different pattern emerges. First, on the town agricultural wage-labour margin (column 1), the night–day opposition disappears ($F = 1.61$, $p = 0.205$): both warmer nights ($+0.063$, SE $= 0.012$) and warmer days ($+0.041$, SE $= 0.016$) raise the peri-urban agricultural wage-labour share, offsetting the -0.05 secular town retreat from farming, while a modest diurnal split remains visible only in the town seasonal residual ($\Delta T_{\min}$: -0.016 vs. $\Delta T_{\max}$: $+0.017$; $F = 4.06$, $p = 0.044$, column 3).

Second—and central to the macroeconomic implications of warming—both nighttime and daytime warming unilaterally contract the non-agricultural worker share in towns (column 4). Against a slight secular rise in town non-agricultural employment ($\Delta\bar{y} = +0.01$), a 1°C warmer night reduces the town non-agricultural share by -0.038 (SE $=$ 0.018) and a 1°C warmer day reduces it by -0.078 (SE $= 0.022$). The two coefficients share the same negative sign and do not statistically separate ($F = 2.78$, $p = 0.096$), implying that both the thermal level and the diurnal range depress urban non-agricultural employment. In the model (Appendix Proposition A7), where agriculture is a minor share of the town economy, direct thermal stress on informal outdoor and physical urban work ($\hat{A}_n < 0$ from daytime heat strain and nighttime recovery loss) dominates, stalling structural transformation at the urban margin in a pattern analogous to the rural-biased town contraction documented by Blakeslee et al. (2023).

Corroborating evidence and robustness. To inspect where heat-displaced urban workers land across detailed industries, the Online Appendix examines urban households across three rounds of the National Sample Survey (1987, 1999, 2009) (Ruggles et al. 2024). The NSS independently reproduces the nighttime rise in peri-urban agricultural wage employment ($+0.060$, $p < 0.01$) and indicates that daytime warming shifts urban workers into informal, low-barrier services—specifically trade, transport, and logistics—rather than formal manufacturing.

The Online Appendix also subjects the baseline estimates to a battery of sensitivity checks: re-estimating the system in decadal panels with district and decade fixed effects and state linear trends; denominating worker counts by total population (per-capita rates immune to the 1991–2001 Census marginal-worker enumeration break); sweeping Conley spatial HAC kernels out to 200 km; and replacing surface-air temperatures with degree-day indices or wet-bulb temperatures. Across specifications, scalar temperature metrics average over the diurnal cycle and mask the opposing rural flows, whereas separating nighttime and daytime warming consistently reveals the rural tug-of-war and the urban non-agricultural drag.

4. Conclusion

This paper asks which way warming temperatures move agricultural labour, and finds that the answer depends on the time of day at which warming occurs. To my knowledge, this paper provides the first reduced-form evidence and a conceptual framework for the economic channels of diurnal warming—shifts in nighttime and daytime temperatures—on structural transformation. The framework assigns the two margins to different factors: nighttime warming is a land-biased shock, daytime warming a factor-neutral one. Over the two-decade long difference this assignment shows in yields, which hold under warmer nights and fall under warmer days ($F = 4.56$ testing $\beta_{\Delta T_{\min}} = \beta_{\Delta T_{\max}}$, Table 1), while cultivated area and the field wage contract by the same amount under either margin and harvest prices do not separate. In the decadal panel the same assignment shows in prices: only the daytime margin raises the local harvest price ($F = 13.34$, online appendix), a medium-run wedge that partial market integration opens and the long difference closes. The framework is a guide to the signs of these comparative statics, which the long-difference baseline and the panel both test. Though the diurnal divergence is robust to a battery of stress tests, the empirical approach has a few important nuances and limitations.

First, while multicollinearity of the within-variation (ρ) is a concern for identification under wet-bulb temperatures ($\rho \approx 0.84$), in the surface-air baseline the two margins are within-correlated at only $\rho \approx -0.09$ (online appendix). Two-way fixed effects and state trends absorb almost all of that co-movement, leaving the identifying variation near-

orthogonal (within-transformed VIF ≈ 1, condition number ≈ 1.1). Both $T_{\min}, T_{\max}$ carry their sign on agricultural labour shares (+/-, respectively) when specified independently, with and without rainfall controls, in both the panel and the long-differenced estimates. The within-correlation widens standard errors without materially altering the magnitude or sign of the point estimates $\beta_{T_{\min}}, \beta_{T_{\max}}$.

Second, the wet-bulb temperature specification is a deterministic nonlinear approximation using the raw dry-bulb $T_{\min}$ and $T_{\max}$ (online appendix), mechanically compressing the diurnal temperature range (DTR). This makes the wet-bulb pair highly collinear relative to the dry-bulb baseline (within-$\rho \approx 0.84$, within-transformed VIF ≈ 3.5; online appendix).

Third, specified jointly, T_{min} and T_{max} span the same two-dimensional regressor space as the daily mean and the diurnal range (DTR). A reparametrisation using the mean-range measures provides supporting evidence that the DTR carries an independent channel on the labour and price margins (online appendix).

Fourth, diurnal warming is a *local* environmental change, compressed or widened by irrigation, aerosols, and land use. These local changes are arguably endogenous to the production function as emphasised by Barrett, Ortiz-Bobea, and Pham (2023). Thus, the diurnal margins are not randomly assigned. Irrigation expansion and other drivers of range compression raise nighttime and lower daytime temperatures in the same agrarian belts where labour reallocates. Three features in the data sample address this confounding: cultivated area contracts identically under both margins, where an irrigation-driven night signal would instead expand it; the nighttime yield dividend persists within the historically irrigated belt (online appendix); and measurement drift in gridded nighttime temperatures near growing settlements would bias the nighttime coefficient toward zero, rejected by the empirical estimates. Moreover, the asymmetry is rooted in atmospheric science: the shallow nocturnal boundary layer amplifies the nighttime response to any change in energy-balance, upstream of local land use (Davy et al. 2016).

Finally, the conceptual framework maintains labour and land as gross substitutes ($\sigma > 1$). Though it is grounded in the agricultural-production literature (Hayami and Ruttan 1985; Binswanger 1974), I do not directly estimate this assumption. The factor bias requires only a non-unit elasticity: under a Cobb–Douglas production function, the land shock folds into neutral productivity (online appendix). The consistent signs on labour, area, output and prices discipline this assignment. Nonlinearity and spatial sorting, held fixed by the linear framework, are left for future research.

The distinction between the two temperature shocks is important for welfare. The nighttime shock halts structural transformation by pulling more hands onto the farm; the daytime shock diverts agricultural wage-employment into a low-productivity, informal

churn within urban services.

A natural extension of the framework is to India's own colonial-era and early post-Independence agricultural records, which provide a test for whether a fully closed subsistence economy without buffers inverts the night-day pattern. The pattern is consistent in itself: warmer nights hold labour on the farm without the price and wage distress that warmer days bring, while daytime heat restrains the non-agricultural transition far more strongly. Given the two margins carry opposite signs on agricultural labour, the trajectory of the diurnal temperature range matters as much as the level. Where nights are warming faster than days—compressing the range (Karl et al. 1991)—the decomposition implies agricultural consolidation and commercialisation. Where days outpace nights and the range widens (Zhong et al. 2023), warming implies the double squeeze of farm-labour casualisation and stalled urban structural transformation.

A back-of-the-envelope calculation—comparing each district's trailing-decade climate at the 2011 and 1981 Censuses—suggests growing-season nights and days warmed by 0.45° and 0.37°C on average over 1981–2011, but the balance is regional—in the districts where the range compressed, nights ran ahead of days (+0.71 versus +0.17°C), and where it widened, days led (+0.54 versus +0.21°C). Applying the long-difference estimates from Table 2, warming in the range-compressing districts implies a 4.3-point rise in the agricultural wage-labour share against an 18% base—a rise of roughly a quarter—and a 4.4-point fall in the casual seasonal share. Warming in the range-widening districts implies a 1.8-point decline in the wage-labour share and a 1.1-point shift onto the seasonal margin. Within the rural workforce, where the divergence is sharpest, the same arithmetic gives +5.7 and −1.7 points. Night and day warming, in equivalent magnitudes, sort local agricultural labour in opposite directions. These findings suggest that under extreme daytime heat, the exit from agricultural labour into informal seasonal employment reflects a physiologically constrained churn for survival, rather than a sustained pathway to structural change.

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Appendix: Model Statements, Summary Statistics, and Supporting Exhibits

This appendix collects the exhibits a reader needs to reproduce the paper's figures and to check the model's predicted signs against the estimates. The full battery of robustness checks, the general-equilibrium model with all proofs, and the inference and measurement diagnostics are in the online appendix.

Model statements cited in the text

Section 1 cites the following objects from the full general-equilibrium model. This subsection reproduces their statements, without proof, so the arXiv working paper is self-contained; all proofs, the closed-economy derivation, and the remaining comparative statics are in the online appendix. Notation ($s_L, s_Z, \varepsilon_a, \tau_a, \eta_a, \chi, \mathcal{M}$) follows Section 1.

ASSUMPTION A1 (Technology and diurnal shocks).

(i) Gross substitutes: $\sigma > 1$.

(ii) Daytime heat lowers TFP: $A'(T_{\max}) < 0$.

(iii) Nighttime warming is land-biased: $B'(T_{\min}) \gtrless 0$, *with* $B'(T_{\min}) \geq 0$ *as the benchmark, corroborated by the positive wet-bulb night yield and the positive long-difference night-minus-day yield wedge (Remark A2).*

LEMMA A1 (Factor-bias decomposition of agricultural labour demand). *To first order, the marginal product of agricultural labour satisfies* $\widehat{\mathrm{MPL}}_a = \hat{A} + \frac{s_Z}{\sigma}(\hat{B} + \hat{Z} - \hat{L}_a)$. *Holding cultivated area fixed* ($\hat{Z} = 0$)*, the farm's labour-demand schedule is equation* (2)*. More generally, at variable area, labour intensity per hectare obeys the same relation:* $\hat{L}_a - \hat{Z} = \hat{B} + \varepsilon_a(\hat{p} + \hat{A} - \hat{w})$.

LEMMA A2 (Observational equivalence under Cobb–Douglas). *If* $\sigma = 1$*, the land-biased shock* $\hat{B}$ *is observationally equivalent to a Hicks-neutral TFP shock scaled by* $s_Z = 1 - \theta$*: agricultural labour demand depends solely on the composite* $\widehat{\mathrm{TFP}} \equiv \hat{A} + (1-\theta)\hat{B}$, $\hat{L}_a = \frac{1}{1-\theta}(\hat{p} + \widehat{\mathrm{TFP}} - \hat{w})$. *Decomposing warming into* $T_{\min}$ *and* $T_{\max}$ *then carries no implication for the relative direction of labour versus output responses.*

LEMMA A3 (Equivalence under a labour-disaugmenting daytime heat shock). *Suppose daytime heat directly impairs human physiological capacity in the field, so effective agricultural labour is* $A_L(T_{\max})L_a$ *with* $A_L'(T_{\max}) < 0$ *and* $\hat{A}_L \equiv \frac{A_L'(T_{\max})}{A_L} dT_{\max} \leq 0$*. Retaining both neutral crop stress* $\hat{A} \leq 0$ *and worker heat stress* $\hat{A}_L \leq 0$*, the farm's labour-demand schedule becomes*

$$\hat{L}_a \;=\; \hat{B} + \varepsilon_a(\hat{p} + \hat{A} - \hat{w}) + \varepsilon_L \hat{A}_L, \qquad \varepsilon_L \equiv \frac{\sigma - s_Z}{s_Z}.$$

Under Assumption A1(i), $\varepsilon_L > 0$*, so direct daytime worker heat stress lowers agricultural labour demand, reinforcing the Hicks-neutral contraction: whether daytime heat is modelled as a pure*

neutral TFP loss, a pure outdoor-worker productivity loss, or a combination, it lowers agricultural labour demand whenever $\sigma > s_Z$, so every sign prediction below is invariant to the day-side assignment.

REMARK A1 (Why nocturnal sleep disruption cannot replace land bias). *The same equation rules out an alternative hypothesis in which warmer nights operate purely by disrupting workers' sleep ($\hat{A}_L(T_{\min}) < 0$) without a land-biased channel ($\hat{B} = 0$). Whenever $\sigma > s_Z$—the exact condition under which daytime heat stress reduces farm labour demand—$\varepsilon_L = (\sigma - s_Z)/s_Z > 0$, so a negative nighttime worker-productivity shock would itself* reduce *agricultural labour demand. Generating the observed nighttime rise in agricultural wage labour therefore requires $\hat{B} > \varepsilon_L |\hat{A}_L(T_{\min})| \geq 0$: land bias must dominate any nighttime fatigue channel.*

REMARK A2 (Empirical evidence on the land-bias sign and yield divergence). *Proposition A2(ii) predicts both a positive night-minus-day yield divergence and, once dry-heat evaporative stress on marginal land is netted out, an outright positive nighttime yield effect. Both are borne out: (1) in the two-decade dry-bulb long difference, the nighttime cereal-yield estimate is slightly positive ($+0.052$) while the daytime estimate is negative (-0.178), rejecting equality at $F = 4.56$ ($p = 0.034$); (2) under humidity-adjusted (wet-bulb) temperatures, which net out dry-air evaporative stress, nighttime warming raises cereal yields outright in both the decadal panel ($+0.15$) and the long difference ($+0.36$) without contracting cultivated area. Because wet-bulb $T_{\min}$, $T_{\max}$ are strongly collinear after fixed effects, the dry-bulb estimates are the cleanly identified baseline and the wet-bulb yield dividend is corroborative evidence of $B'(T_{\min}) > 0$.*

PROPOSITION A1 (Common cultivated-area contraction). *Both nighttime and daytime warming contract cultivated area: $\partial \ln Z / \partial T_{\min} \approx \partial \ln Z / \partial T_{\max} < 0$. Equality of the two semi-elasticities corresponds to a calibration of the extensive marginal-cost shifters, consistent with the empirical failure to reject equality of the area coefficients in both the decadal panel ($F = 0.01$) and the long difference ($F = 0.03$).*

PROPOSITION A2 (Decoupling of labour and yield under nighttime warming). *In the open economy: (i) the sign of the nighttime agricultural labour response is pinned down solely by the sign of the land-bias shock, $\operatorname{sgn}(\partial \hat{L}_a / \partial T_{\min}) = \operatorname{sgn}(B'(T_{\min}))$, attenuated by $\tau_a \in (0, 1)$; (ii) the crop-yield response $\hat{y}_a = \hat{A} + s_L(\hat{L}_a - \hat{Z}) + s_Z \hat{B}$ is weakly positive under nighttime warming and strictly exceeds the daytime yield response whenever $B'(T_{\min}) \geq 0 > A'(T_{\max})$; (iii) under daytime warming, labour, output, and yield all move in the same negative direction.*

LEMMA A4 (Counter-flows between wage labour, self-cultivation, and seasonal work). *When district non-agricultural employment shares remain flat ($d\ell_n \approx 0$), changes in the agricultural wage-labour share $d\ell_a = \ell_a \hat{L}_a$ are exactly offset by the sum of self-cultivation and seasonal shares: $d\ell_c + d\ell_s = -d\ell_a = -\ell_a \hat{L}_a$. Under nighttime warming ($\hat{L}_a > 0$), expanding farm wage labour draws workers out of both self-cultivation and the seasonal pool. Under daytime warming ($\hat{L}_a < 0$), displaced wage workers are absorbed primarily by the elastic seasonal buffer.*

PROPOSITION A3 (Diurnal labour divergence in the open economy). *Under Assumption A1, agricultural wage labour rises with nighttime and falls with daytime warming:* $\partial \hat{L}_a / \partial T_{\min} = \tau_a \cdot B'(T_{\min})/B \geq 0$, $\partial \hat{L}_a / \partial T_{\max} = \tau_a \varepsilon_a \cdot A'(T_{\max})/A < 0$, *with strictly opposite signs whenever* $B'(T_{\min}) > 0$ *and* $A'(T_{\max}) < 0$. *The seasonal labour pool moves oppositely to agricultural wage labour.*

PROPOSITION A4 (Sign reversal in the closed subsistence economy). *In a fully closed subsistence economy (*$\chi = 1$*) with* $\mathcal{M} > 1$ *and* $\sigma > 1$, *both diurnal labour signs reverse:* $\partial \hat{L}_a / \partial T_{\min} < 0$ *and* $\partial \hat{L}_a / \partial T_{\max} > 0$*—the exact opposite of the open economy and of the Indian data.*

PROPOSITION A5 (Openness threshold for the observed sign orientation). *Under Assumption A1 with* $B'(T_{\min}) > 0$, *the equilibrium exhibits the empirically observed orientation—nighttime warming raising and daytime warming lowering agricultural wage labour—if and only if local food-market closure lies below the threshold* $\chi < \chi^* \equiv 1/(\sigma \mathcal{M})$. *Because* $\sigma > 1$, *the nighttime condition is strictly more binding than the daytime condition (*$\chi < 1/\mathcal{M}$*).*

REMARK A3 (Marketed surplus versus subsistence marginal parcels). *Why does nighttime cultivated-area abandonment in the decadal panel not raise the local harvest price alongside daytime warming? Nighttime land abandonment concentrates on marginal self-cultivated plots whose output is consumed on-farm: when a marginal cultivator hires out as a wage labourer, household subsistence demand and supply move together, leaving marketed surplus on commercial inframarginal land buffered by higher hired-labour intensity. Daytime heat, by contrast, reduces productivity across all inframarginal marketed grain, tightening local marketed supply and raising the harvest price whenever* $\chi > 0$. *Under wet-bulb temperatures, where nighttime warming raises inframarginal yields outright without area loss, marketed supply expands and the nighttime harvest price falls significantly, confirming that the price closure operates on marketed surplus.*

PROPOSITION A6 (Rural amplification of the diurnal labour divergence). *For a given climate shock, the change in the agricultural wage-labour share* $d\ell_a = \ell_a \hat{L}_a$ *is strictly larger in magnitude in rural than in urban populations, driven by two reinforcing channels: (i)* baseline agricultural scale*—the initial share* ℓ_a *is several times larger in rural populations, scaling any given percentage response into a proportionally larger percentage-point share change; (ii)* depth of the seasonal labour buffer*—the casual seasonal supply elasticity* η_s *is deeper in rural areas, raising* $\eta_a = \eta_s + \phi_n \varepsilon_n$ *and pushing the incidence ratio* $\tau_a = \eta_a/(\eta_a + \phi_a \varepsilon_a)$ *toward* 1.

PROPOSITION A7 (Unilateral non-agricultural contraction in towns). *Suppose non-agricultural productivity falls with both nighttime and daytime warming. Then in an open town economy (*$\chi \approx 0$*) where agriculture is a minor employment share (*ϕ_a *small), non-agricultural employment contracts under both margins:* $\partial \hat{L}_n / \partial T_{\min} < 0$ *and* $\partial \hat{L}_n / \partial T_{\max} < 0$.

TABLE A1. Model comparative statics: predicted signs and their empirical counterparts

	Predicted sign		
Outcome	T_{min} (night)	T_{max} (day)	Empirical counterpart
Cultivated area	$(-)$	$(-)$	Equal contraction; $F = 0.01$ (Table 1)
Harvest price	(≈ 0)	$(+)$	Only daytime raises price; $F = 13.3$ (Table 1)
Field wage	(≈ 0)	$(-)$	Daytime wage loss (Table 1)
Cereal yield	$(\approx 0, +)$	$(\approx 0, -)$	Flat in dry-bulb; $+0.15$ wet-bulb night (online app.)
Ag. wage-labour share	$(+)$	$(-)$	Night raises, day lowers; $F = 24.9$ (Table 2)
Seasonal share	$(-)$	$(+)$	Opposite of wage labour; $F = 14.7$ (Table 2)
Cultivator share	$(-)$	(≈ 0)	Drawn off at night (Table 2)
Rural vs. urban amplitude	same sign, rural $>$ urban		$F = 30.9$ vs. 1.9 (Appendix Table A3)
Towns (non-agriculture)	$(-)$	$(-)$	Both margins depress; $F = 2.8$ (Table 2, Panel B)
Closed economy	signs *reverse*		Not observed $\Rightarrow$ open ($\chi < \chi^*$)

Notes: Signs are the framework's predictions in the empirically selected partial-open regime $0 < \chi < \chi^*$ under gross substitutes ($\sigma > 1$); the price asymmetry is the over-identifying sign restriction and the area equality is a calibration rather than a theorem. Only signs and the ordering of magnitudes are predicted, not levels. The full model, proofs, and the wet-bulb land-augmentation evidence are in the online appendix.

TABLE A2. Summary statistics: district and town, panel and long-difference samples

	Decadal panel (levels) 1981–2011		Long difference (Δ 2001–11 vs. 1981–91)	
	District	Town	District	Town
Cultivators (share)	0.323	0.080	-0.139	-0.058
	(0.158)	(0.091)	(0.080)	(0.057)
Agricultural labourers (share)	0.180	0.088	-0.049	-0.053
	(0.111)	(0.097)	(0.057)	(0.060)
Seasonal / marginal (share)	0.173	0.085	0.139	0.102
	(0.111)	(0.079)	(0.086)	(0.065)
Non-agricultural (share)	0.324	0.745	0.048	0.010
	(0.176)	(0.178)	(0.070)	(0.089)
Nighttime T_{min} (°C)	17.95	18.23	0.28	0.29
	(2.39)	(2.22)	(0.24)	(0.24)
Daytime T_{max} (°C)	29.71	29.89	0.15	0.11
	(1.95)	(1.63)	(0.19)	(0.20)
Rainfall (growing-season mean)	1.0	1.0	0.0	0.0
	(0.5)	(0.5)	(0.1)	(0.1)
Units (districts / towns)	300	2,969	299	2,969
Observations	1,187	11,876	299	2,969

Notes: Means with standard deviations in parentheses, for the decadal panel (levels, 1981–2011) and the long-difference cross-section (Δ 2001–11 vs. 1981–91), each split into the district (total-population) and Census-town samples. Worker shares are of total workers in the unit; temperature and rainfall are growing-season ten-year moving averages. The town columns use the balanced panel of towns present in every round since 1981—the long-difference sample—so the panel-levels and long-difference town statistics describe the same 2,969 towns. The district panel counts one more unit than the long difference (300 vs. 299): one district is observed only at the endline and so enters the panel but cannot be differenced.

TABLE A3. District worker reallocation, long difference (1981–2011), by residence

	Share of Workers				
	Ag. Labour	Cultivator	Seasonal	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
Panel A. Total population					
ΔT_{min}	**0.075**	-0.024	**-0.074**	0.026	0.297
	(3.58)	(-0.79)	(-2.74)	(0.94)	(1.24)
ΔT_{max}	**-0.062**	-0.008	0.049	0.020	0.041
	(-2.51)	(-0.22)	(1.47)	(0.55)	(0.16)
F ($\Delta T_{min} = \Delta T_{max}$)	26.07	0.16	12.34	0.03	0.83
p-value	<0.001	0.687	<0.001	0.865	0.363
Observations	293	293	293	293	293
Panel B. Rural population					
ΔT_{min}	**0.096**	-0.035	**-0.064**	0.007	0.211
	(4.12)	(-1.11)	(-2.25)	(0.37)	(0.91)
ΔT_{max}	**-0.068**	0.011	0.056	-0.001	0.025
	(-2.52)	(0.28)	(1.58)	(-0.02)	(0.10)
F ($\Delta T_{min} = \Delta T_{max}$)	30.87	1.24	11.36	0.07	0.48
p-value	<0.001	0.266	<0.001	0.794	0.489
Observations	291	291	291	291	293
Panel C. Urban population					
ΔT_{min}	**0.034**	-0.013	**-0.028**	0.007	0.632
	(2.64)	(-0.89)	(-2.09)	(0.25)	(1.75)
ΔT_{max}	0.006	0.015	0.006	-0.028	0.305
	(0.32)	(0.89)	(0.37)	(-0.75)	(0.80)
F ($\Delta T_{min} = \Delta T_{max}$)	1.89	1.69	3.23	0.69	0.60
p-value	0.171	0.194	0.073	0.407	0.439
Observations	291	291	291	291	293
Rainfall Control	Y	Y	Y	Y	Y
State FE	Y	Y	Y	Y	Y

Notes: Long-difference estimates of nighttime (ΔT_{min}) and daytime (ΔT_{max}) growing-season surface-air warming on district worker shares, decomposed into the total, rural, and urban populations of each district (the within-district residence split): Panel A is the total-population estimate discussed in the main text; Panel B is the rural residence split behind Table 2, Panel A; Panel C is the urban residence split discussed alongside it. Point estimates with $|t| > 2$ in bold over district-clustered t-statistics; the F-statistic and its p-value test $\Delta T_{min} = \Delta T_{max}$. All specifications include state fixed effects and a precipitation control.

TABLE A4. Town worker reallocation, long difference (1981–2011)

	Share of Workers				
	Ag. Labour	Cultivator	Seasonal	Non-Agriculture	Log Total Workers
	(1)	(2)	(3)	(4)	(5)
ΔT_{min}	**0.063**	-0.009	-0.016	**-0.038**	0.272
	(5.46)	(-0.86)	(-1.35)	(-2.16)	(1.61)
ΔT_{max}	**0.041**	0.022	0.017	**-0.078**	0.241
	(2.59)	(1.49)	(1.15)	(-3.47)	(1.41)
Observations	2,969	2,969	2,969	2,969	2,969
F-stat ($\Delta T_{min} = \Delta T_{max}$)	1.61	3.38	4.06	2.78	0.11
p-value	0.205	0.067	0.044	0.096	0.738
Rainfall Control	Y	Y	Y	Y	Y
FE / Trend	Y	Y	Y	Y	Y

Notes: Census-town worker reallocation under nighttime (ΔT_{min}) and daytime (ΔT_{max}) growing-season surface-air warming, across the balanced panel of Census towns—the coefficients behind Table 2, Panel B. The four mutually exclusive worker categories (agricultural wage labour, cultivators, seasonal, non-agricultural) sum to one, so the share coefficients sum to approximately zero; column 5, set apart, reports the long-run change in log total town workers. Point estimates with $|t| > 2$ in bold over district-clustered t-statistics; the F-statistic and its p-value test $\Delta T_{min} = \Delta T_{max}$. State fixed effects and a precipitation control throughout.